Shubham Gaur

Santa Clara, CA, USA • gshubhaur@gmail.com
github.com/shubham2345 • linkedin.com/in/shubhamggaur • shubham2345.github.io • Google Scholar

Summary

First-year Ph.D. student at UC Santa Cruz (advisor: Dr. Ian Lane) researching LLM post-training, factuality, and safety-focused evaluation of LLMs and agents. Work spans agent benchmarks, reward hacking in agent evaluation, and user dependence on supportive LLMs, with a co-first-author ICLR 2027 submission and 1st place at UC Berkeley AgentBeats 2026. Brings 6 years of production ML engineering at 100M+ user scale.

Education

Ph.D., Computer Science • University of California, Santa Cruz Sep 2026 – Present (expected Dec 2029)
Advisor: Dr. Ian Lane • Research: LLM Post-Training, Factuality, Human-Centered Alignment

M.S., Natural Language Processing (AI) • University of California, Santa Cruz • GPA 4.0/4.0 Sep 2024 – Mar 2026
Advisor: Dr. Chenguang Wang • Research: LLM Agents, Benchmarks & Evaluation

B.S., Computer Science • SRM Institute of Science & Technology, Chennai • GPA 3.8/4.0 Jul 2015 – May 2019
Advisor: Dr. A. Shanthini • Research: Recommendation Systems, Computer Vision, Visual Question Answering

Publications & Preprints

*Equal contribution (co-first author).

[1] Shukla, S.*, Kim, J.*, **Gaur, S.***, Roshanaei, M., & Seif El-Nasr, M. (2026). RELATE: An Evaluation Framework for Measuring Relational Orientation of Large Language Models. *Under review at ICLR 2027.* arXiv:2610.09569.

[2] Zheng, S., Di, Z., Liu, Y., . . . , **Gaur, S.**, Ye, Z., Wang, H., Li, X., Song, D., & Hauser, C. (2026). BenchShield: Formal Model-Backed Instrumentation for Reward Integrity in LLM-Agent Evaluation Infrastructure. *Preprint.* arXiv:2609.11028.

[3] Yagnik, S, **Gaur, S.**, Thakur, S., Jain, V., Chadha, A., & Das, A. (2026). MAAT: Multi-phase Adapter-Aware Targeted Unlearning. *Preprint.* arXiv:2605.30514.

[4] **Gaur, S.**, & Lane, I. (2026). Signal-Driven Observation for Long-Horizon Web Agents. *ICML 2026 Workshop on Failure Modes in Agentic AI (FAGEN), Position Paper.* arXiv:2606.06708.

[5] Crispino, N., **Gaur, S.**, Yang, X., Yu, A., Ercevik, B., Sapugay, C., Potter, Y., Song, D., & Wang, C. (2026). SkillOptimizer: Agent Skill Optimization Through Subskills Without Task Supervision. *ICML 2026 AIWILD: Safety, Security, and Beyond Workshop.*

[6] Tu, J., **Gaur, S.**, Murtinty, R., Wang, Z., Shi, T., Song, D., & Wang, C. (2026). FAULTLOC: Evaluating AI Coding Agents for Fault Localization from Crash to Cause. *ICML 2026 AIWILD: Safety, Security, and Beyond Workshop.*

[7] Clymo, J., Zernik, A., & **Gaur, S.** (2025). UCSC NLP T6 at SemEval-2025 Task 1: Leveraging LLMs and VLMs for Idiomatic Understanding. *Proceedings of SemEval-2025 (ACL Workshop).*

[8] Kilaru, S., **Gaur, S.**, & Rout, S. (2024). Advancing Web-Based Visual Question Answering with Efficient Image-Text Alignment. *Recent Advancements in Artificial Intelligence: Proceedings of ICRAAI 2023*, pp. 343–359. Springer.

Research Experience

Graduate Researcher • UC Santa Cruz • Santa Cruz, CA Apr 2026 – Present
Advisor: Dr. Ian Lane • Pre-doctoral researcher; Ph.D. student since Sep 2026

- Investigating long-tail factual recall in LLMs via reasoning and post-training.
- Defined *observation over-ingestion* as a root cause of context rot and goal drift in long-horizon web agents; proposed Signal-Driven Observation, re-reading the DOM only on browser-event signals (first author, ICML 2026 FAGEN position paper).

Graduate Research Assistant • Wang's Research Lab • Santa Clara, CA Oct 2025 – Aug 2026
Advisor: Dr. Chenguang Wang • In collaboration with UC Berkeley RDI (Prof. Dawn Song)

- Designed FAULTLOC, a 900-CVE benchmark evaluating AI coding agents on fault localization; ran ablations across frontier models, showing frontier agents recover the true root cause in only 40% of cases.
- Built RCABench, benchmark infrastructure for multi-hop causal reasoning and root-cause localization over large codebases; surfaced systematic multi-agent coordination failures under realistic security conditions.
- Developed SkillOptimizer, improving LLM agent skills without task supervision via subskill DAG decomposition and coverage-driven optimization; raised task pass rates across 192 skills and three agent families.
- Contributed to MassGen, an open-source **multi-agent LLM framework** for coordinating heterogeneous models on complex, open-ended tasks.

Open-Source Contributor • BenchFlow Jul 2026 – Present

- Adjudicated reward-hacking trajectories for 35 tasks against a structured rubric for BenchShield, contributing to a 456-trajectory human-adjudicated corpus drawn from 31,000+ agent runs (co-author).
- Contributing to FrontierPhysics, a benchmark of specialist-level physics research tasks: surveyed physics-evaluation literature, built a platform connecting the project with Nobel laureate physicists, and fixed bugs in BenchFlow's evaluation infrastructure.

Cross-Institution Research Collaborations 2026

- RELATE (with UCSC GUII Lab and researchers at Stanford): Co-developed a framework measuring whether supportive LLMs foster reliance on the AI or encourage human connection; across 1,596 dialogues and 7 LLMs, AI-centered language rose from 40.5% to 72.9% over six turns (co-first author, under review at ICLR 2027).
- MAAT (with researchers at Stanford and BITS Pilani): Co-developed a three-phase LoRA-adapter unlearning method and 5WBench, a benchmark for causal-knowledge unlearning; MAAT retained 36.4 points more knowledge than the strongest baseline at equal forgetting.

Industry Experience

Applied Research Intern • Nokia Bell Labs • Naperville, IL, USA Jun 2025 – Aug 2025

- Designed a multimodal retrieval pipeline for enterprise knowledge search with vector databases (FAISS, Chroma); evaluated retrieval latency and ranking quality with custom metrics and A/B tests.
- Built transformer-based time-series forecasting and graph neural network anomaly detection; optimized serving via GPU acceleration and ONNX

conversion.

Founding Lead Machine Learning Engineer • **Adani Group** • **Gurgaon, India** Nov 2021 – Aug 2024

- Architected ML infrastructure for a 100M+ user personalization system; ran controlled A/B experiments across 50+ model variants, improving engagement by 35%.
- Built an end-to-end PyTorch training and deployment pipeline with data/model drift monitoring and automated retraining, cutting model refresh time by 60%.
- Part of the team that won the MongoDB APAC Innovation Award 2023 (From Batch to Real-Time) for re-architecting the AdaniOne SuperApp data platform from batch to real-time event streaming.

Machine Learning Engineer • **BlackRock** • **Mumbai, India** Jul 2019 – Oct 2021

- Built deep neural networks for fixed-income and equity portfolio optimization, cutting inference latency 60% via quantization and pruning.
- Developed risk-scoring ensembles with SHAP explainability and statistically significance-tested model comparisons.

Skills

- **Languages:** Python, C++, SQL, Java
- **ML Frameworks:** PyTorch, JAX, Hugging Face Transformers, TensorFlow, scikit-learn
- **LLM Training & Post-Training:** Supervised Fine-Tuning (SFT), Reinforcement Learning from Human Feedback (RLHF), Direct Preference Optimization (DPO), LoRA/PEFT, Machine Unlearning, Knowledge Distillation
- **LLM Agents & Evaluation:** Agent Benchmarks, LLM-as-a-Judge, Reward Hacking Analysis, Multi-Agent Systems, Human Evaluation, Ablation Studies, Statistical Significance Testing
- **Systems & Infrastructure:** Distributed Training, CUDA, Triton, ONNX, Ray, Docker, Kubernetes, Spark, AWS SageMaker, GCP Vertex AI

Honors & Awards

- Recipient, UC Santa Cruz Regents Award and International Doctoral Recruitment Fellowship (2026).
- Winner, NVIDIA GTC 2026 PyTorch Helion Hackathon (LLM-guided Triton kernel optimization)
- 1st Place, Cybersecurity Agent track, UC Berkeley AgentBeats Competition 2026.
- BlueDot Impact AI Safety Fellowship (2026), a selective cohort-based AI alignment program..
- NVIDIA Certified Associate & Professional: Generative AI LLMs
- Recipient, Thinking Machines Lab Tinker Research Grant (2025), with Dr. Chenguang Wang.
- Peer Reviewer: ICLR 2026, ACL 2025.